

CE2F4B105/6
CONSTRUCTION ENGINEERING & MANAGEMENT
IIT MADRAS

CAPSTONE REPORT

Residential Building Design & Construction Planning

250 sq.m. Single-Family Home – Hyderabad

GROUP - 2

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1. Project Overview & Floor Plan

This report presents a comprehensive construction plan for a 250 sq.m. ($\approx 2,690$ sq.ft.) single-storey residential building designed for Hyderabad's climate conditions. The project has been planned in compliance with GHMC (Greater Hyderabad Municipal Corporation) building by-laws, IS 456:2007 for reinforced concrete design, and NBC 2016 standards.

1.1 Site & Plot Details

Parameter	Details
Plot Area	2,592 sq.ft $\approx$ 241 sq.m (37 ft $\times$ 70 ft approx.)
Built-up Area	250 sq.m (Ground Floor)
Location	Hyderabad, Telangana (Tropical Wet & Dry climate zone)
Design Life	50–60 years (per IS 456:2007)
Building Regulation	GHMC Building Laws; NBC 2016 Parts 3 & 8
Structural System	RCC Frame Structure – Isolated Column Footings
Scale (Drawing)	1 cm = 2.7 ft

1.2 Room Schedule & Areas

#	Room	Size (ft)	Area (sq.ft)	Area (sq.m)
1	Living Room	10 $\times$ 35	350	32.5
2	Master Bedroom	13 $\times$ 15	195	18.1
3	Kids Bedroom	12 $\times$ 15	180	16.7
4	Guest Room	12 $\times$ 10	120	11.1
5	Family Lounge	13 $\times$ 12	156	14.5
6	Kitchen	9 $\times$ 11	99	9.2
7	Entry Foyer	7 $\times$ 11	77	7.2
8	Dining Hall	10 $\times$ 9	90	8.4
9	Playing Area	28 $\times$ 4 (open)	112	10.4
10	Walk-in Closet	8 $\times$ 4	32	3.0
11	Bathroom	8 $\times$ 4	32	3.0
12	Attached Toilet	10 $\times$ 4	40	3.7
13	Utility / Wash	4 $\times$ 3	12	1.1
14	Rear Private Garden	7 $\times$ 35	245	22.8

1 5	Parking (Open)	15 × 36	540	50.2
1 6	Pooja Area	–	~30	~2.8
	Circulation Hallway	–	~120	~11.1
	TOTAL BUILT-UP		2,690 sq.ft	250 sq.m

1.3 Climate-Responsive Design Principles

Hyderabad experiences a Tropical Wet & Dry climate. The primary challenges are intense heat and maintaining thermal comfort without excessive air conditioning. The following design strategies have been incorporated:

- Service zones (toilets, utility room) and the Rear Private Garden are positioned as thermal and acoustic buffers, shielding habitable rooms from heat and street noise.
- The Central Circulation Hallway promotes cross-ventilation; airflow enters from the front and exits through rear openings when windows are open.
- Bedrooms are placed at the rear to isolate them from street noise, maintain privacy, and benefit from the garden's thermal buffer.
- A large roof area of ~250 sq.m is designed to facilitate rainwater harvesting (RWH), providing space for a stormwater tank to address Hyderabad's water scarcity.
- Windows area is designed to be $\geq 1/6$ th of floor area as required by NBC 2016 Part 8.
- Open parking with a cantilever shade structure replaces a full garage to save cost and space.
- Dining acts as the central hub connecting the kitchen, living room, hallway, and main entry.
- Functional zoning separates the public zone (living, dining, foyer) at the front from the private zone (bedrooms) at the rear.
- Two-door toilet arrangement creates dual-access utility, saving ~40 sq.ft that would be consumed by an additional independent toilet.

2. Materials Selection – Justification & Characteristics

All materials have been selected considering Hyderabad's hot climate, NBC 2016 and IS code compliance, lifecycle cost, sustainability, and local availability.

2.1 Foundation

Aspect	Details	Applicable Standard
Material	Reinforced Concrete (M25 grade) – Standard Isolated Column Footings	IS 456:2007
Justification	Most cost-effective for a single-storey residential structure; suitable for Hyderabad's hard-rock and red-soil strata; provides adequate bearing capacity	–
Key Properties	High compressive strength (≥ 25 N/mm ²); durability; resistance to seismic loads; ductility provided by steel reinforcement	–
Tests Required	Compressive strength test (cube test), tensile strength test, slump test, shear test, chemical test (to ensure neutrality – no corrosion of rebar), ductility test	IS 516, IS 1199

2.2 Walls

Aspect	Details	Standard
Material	AAC Blocks – Autoclaved Aerated Concrete (external: 200 mm thick; internal: 150 mm thick)	IS 2185-Part 3
Justification	Superior to traditional red clay bricks for Hyderabad's heat. Excellent thermal insulation reduces air-conditioning load by 20–30%. Lightweight (reduces dead load on structure). Sustainable – lower embodied energy than clay bricks.	–
Key Properties	Thermal conductivity: 0.16–0.18 W/mK (vs 0.81 for clay brick); Density: 550–650 kg/m ³ ; Fire-resistant up to 4 hours; Low porosity; Good acoustic insulation	–
Tests Required	Compressive strength test, passive porosity test, density test, shrinkage test, shear strength test	IS 2185

2.3 Roof

Aspect	Details	Standard
Material	RCC Slab (M25, 125 mm thick) + Cool Roof Coating + PVC Sheet Waterproofing Layer	IS 456:2007
Justification	RCC provides structural integrity and supports the large roof area for RWH. Cool roof reflective coating reduces indoor temperature by 2–5°C in Hyderabad's summers. PVC layer prevents water seepage during monsoon.	–

Key Properties	High compressive strength; Solar Reflectance Index (SRI) > 78 for cool coat; PVC: 20+ year lifespan, UV-resistant; overall system reduces cooling loads by ~25%	–
Tests Required	Compressive strength test, tensile strength test, corrosion test (for rebar), porosity test, water-tightness test	IS 516, IS 3025

2.4 Flooring

Zone	Material	Justification & Properties	Tests
All rooms except Living & Kids Bed.	Kota Stone	Locally sourced from Rajasthan (cost-effective via Hyderabad supply); non-porous; very durable; cool to touch (low thermal conductivity – ideal for barefoot walking in heat); sustainable	Reactivity, thermal conductivity, friction test
Living Room & Kids Bedroom	Engineered Wood	High strength-to-weight ratio; excellent aesthetic appeal; good thermal comfort; longevity 20+ years; sustainable with FSC-certified wood	Density, strength, friction, hardness, thermal expansion
Kitchen & Toilets	Granite (floor) + Ceramic Tiles (wall)	Granite: hard, scratch-resistant, non-porous, hygienic, moisture-resistant. Ceramic tiles: easy to clean, protects walls from water damage, low maintenance.	Porosity, friction, hardness test

2.5 Material Tests Summary (Q6 Answer)

Material	Key Tests	Why
Concrete (M25)	1) Compressive strength (cube test)2) Flexural strength3) Chemical test (neutrality – no rebar corrosion)4) Slump test (workability)5) Shear test	Ensures design strength, workability, and long-term durability of structural elements
AAC Blocks	1) Compressive strength2) Passive porosity3) Density & shrinkage test4) Shear strength test	Verifies thermal performance and structural adequacy per IS 2185
Steel Rebar	1) Tensile strength2) Flexural strength3) Thermal expansion coefficient4) Reactivity / corrosion resistance5) Ductility	Ensures rebar can handle tension loads and won't corrode in Hyderabad's humid conditions
Ceramics / Granite	1) Reactivity2) Porosity3) Strength4) Friction5) Thermal conductivity	Ensures hygiene, slip resistance, and durability in wet zones
Glass (Windows)	1) Tensile strength2) Sustainability / recyclability	Safety glazing and lifecycle assessment
Wood / Kota Stone	1) Strength2) Friction3) Hardness4) Thermal expansion	Ensures dimensional stability and comfort under Hyderabad's temperature swings

3. Structural Details

3.1 Foundation Details

Type: Isolated Reinforced Concrete Column Footings (Standard Isolated Footings)

Parameter	Value
Number of columns / footings	10 columns (approximately, based on plan grid)
Footing plan size	1,500 mm × 1,500 mm (stepped isolated)
Footing depth	1,000 mm below GL (assumes founding level at hard stratum)
Footing thickness	300 mm (bottom slab)
Grade of concrete	M25 (fck = 25 N/mm ²) per IS 456:2007
Cover (clear)	50 mm (for footing in contact with soil)
Main reinforcement	12 mm dia HYSD bars @ 150 mm c/c (both directions)
PCC bed (levelling)	75 mm thick 1:4:8 PCC below footing
Plinth beam	230 mm × 300 mm RCC beam connecting all columns at plinth level
Plinth height	600 mm above GL (damp-proof course at plinth level)
Soil bearing capacity assumed	150 kN/m ² (safe bearing capacity for hard murrum)

3.2 Structural Frame

System: Reinforced Concrete Frame – Beams, Columns & Slab per IS 456:2007

Element	Specification
Columns	230 mm × 230 mm; M25 concrete; 4 nos. 12 mm dia HYSD bars; lateral ties 8 mm dia @ 150 mm c/c
Floor height	3,000 mm (floor to floor)
Main beams	230 mm × 450 mm deep; M25; 3 nos. 16 mm dia bars (tension); 2 nos. 12 mm (compression); stirrups 8 mm @ 150 mm
Secondary / tie beams	230 mm × 300 mm; same concrete grade
Roof slab	125 mm thick flat RCC slab; M25; 10 mm dia HYSD @ 150 mm c/c both ways; effective span up to 4 m
Steel grade	Fe 415 HYSD (TMT bars) per IS 1786
Concrete mix design	M25 per IS 10262:2019; w/c ratio ≤ 0.50
Cover – beams/columns	40 mm clear cover
Cover – slab	20 mm clear cover
Seismic zone	Zone II (Hyderabad) – design per IS 1893:2016

3.3 Roof

- RCC flat slab, M25, 125 mm thick, spanning between beams.
- Cool roof reflective coating (SRI > 78) applied over slab to reduce surface temperature.
- PVC waterproofing membrane (1.5 mm thick) installed over sloped screed to prevent seepage.
- Roof slope: 1% towards drains for RWH collection.
- Estimated roof water yield ($250 \text{ sq.m} \times 800 \text{ mm annual rainfall} \times 0.85 \text{ coefficient}$) $\approx 170,000$ litres/year.
- Underground sump tank (15,000 litres) for stormwater storage.

3.4 Walls

- External walls: 200 mm AAC block masonry in 1:6 cement-sand mortar.
- Internal partition walls: 150 mm AAC block masonry.
- All external surfaces: 12 mm thick cement-sand plaster (1:4) + external weather-coat paint.
- All internal surfaces: 12 mm thick gypsum or cement plaster (1:6) + interior emulsion paint.
- Damp Proof Course (DPC): 20 mm thick cement concrete 1:1.5:3 at plinth level, waterproofed with Dr. Fixit or equivalent.
- Window sills, lintels, and chajjas: RCC 100 mm thick, M20 concrete.

4. Construction Equipment Required

The following equipment is required for the construction of the 250 sq.m. residential building:

#	Equipment	Quantity	Duration of Use	Purpose
1	JCB / Mini Excavator (0.2 m ³ bucket)	1 No.	3–5 days	Excavation of column pits and plinth trenches
2	Concrete Mixer (350 L drum)	1 No.	35 days	Mixing concrete for foundations, columns, beams, and slab
3	Needle Vibrator (25 mm & 40 mm)	2 Nos.	35 days	Compaction / consolidation of concrete to remove voids
4	Bar Bending Machine	1 No.	20 days	Cutting and bending steel reinforcement bars to required shapes
5	Shuttering / Formwork (steel plates)	Full set	30 days	Formwork for columns, beams, slabs, and lintels
6	Transit Mixer / Ready Mix Concrete Truck	As needed	4–5 visits	Supply of RMC for roof slab pour (bulk quantity)
7	Concrete Pump	1 No.	2 days	Placing roof slab concrete efficiently over large area
8	Tower Crane / Hydraulic Crane (5T)	1 No.	15 days	Lifting heavy materials – steel, precast elements, formwork
9	Welding Machine	1 No.	10 days	Welding reinforcement chairs and structural steel connections
10	Scaffolding (cup-lock type)	Full set	25 days	Working platform for masonry, plastering, painting at height
11	Water Pump (submersible)	1 No.	Throughout	Water supply for curing concrete and masonry
12	Dumper / Tipper (3T)	1 No.	5 days	Removal of excavated material from site
13	Plate Compactor	1 No.	3 days	Compaction of backfill soil around foundations
14	Hand tools (trowels, plumb bobs, spirit levels, mason's line)	Full set	Throughout	Masonry, plastering, finishing, and surveying
15	Total Station / Theodolite	1 No.	2 days	Setting out building lines, column centre-lines, checking levels
16	Batching Equipment (weigh batching)	1 No.	20 days	Accurate proportioning of concrete ingredients
17	Cube Moulds (150 mm)	12 Nos.	Throughout	Quality control cube casting for concrete testing
18	Slump Cone Test Set	2 sets	Throughout	Workability testing of fresh concrete at site

5. Building Services

The following services are planned for the residential building, designed to comply with NBC 2016 and relevant IS codes.

5.1 Water Supply & Plumbing

- Source: GHMC municipal water supply + overhead tank (2,000 litres) + underground sump (15,000 litres for RWH).
- Pipe material: CPVC pipes (hot water) and UPVC pipes (cold water), ISI marked.
- Dedicated wet areas: 2 bathrooms, 1 attached toilet, utility/wash area, kitchen.
- Solar water heater (200 LPD) on roof for hot water supply to bathrooms and kitchen.
- Water meter provided at main entry; pressure-reducing valve to maintain 1–3 bar at fixtures.

5.2 Drainage & Sanitation

- Waste water from kitchen and utility: connected to external drainage via grease trap.
- Sewage from toilets: connected to septic tank (2,000 L capacity) + soak pit, or GHMC sewer connection.
- Storm water from roof and open areas: channelled to underground sump and overflow to street drain.
- Pipe material: UPVC / SWR (Soil-Waste-Rain) pipes; 110 mm for soil pipes, 75 mm for waste pipes.
- Inspection chambers at junctions; floor traps with water seals in all wet areas.

5.3 Electrical Installation

- Supply: 3-phase, 415 V DISCOM connection; consumer DB with MCBs and RCCBs per IE Rules 2023.
- Estimated connected load: ~8 kW (lighting, fans, appliances); sanctioned load applied through TSSPDCL.
- Wiring: FR (Flame Retardant) copper wires in concealed PVC conduit; 2.5 sq.mm for points, 4 sq.mm for A/C.
- Earthing: Pipe earthing per IS 3043; separate earth leakage circuit breaker for wet zones.
- Solar PV provision: 2 kWp rooftop solar provision with net-metering connection.
- External lighting: driveway and garden LED fittings with automatic dusk-to-dawn sensor.

5.4 HVAC (Ventilation & Cooling)

- Natural ventilation through cross-ventilation via Central Hallway; no mechanical HVAC in primary design.
- Ceiling fans in all rooms (750 mm sweep) as primary cooling.
- Split AC provision (1.5 TR) in master bedroom and kids bedroom; concealed conduit for refrigerant piping.
- AAC block walls + cool roof + thermal mass significantly reduce cooling load.
- Exhaust fan (150 mm) in kitchen, bathrooms, and utility room per NBC requirements.

5.5 Fire Protection

- 2 nos. ABC dry powder fire extinguishers (2 kg) – one in kitchen, one near entrance.
- Smoke detectors in all bedrooms and living room; heat detector in kitchen.

- Fire-resistant AAC block walls provide >4-hour fire rating (no additional fire treatment needed).
- Clear 3 m setback on all sides maintained per GHMC for fire access.

5.6 Communication & IT

- Telephone / broadband: conduit provision through external wall for OFC fibre connection.
- CCTV provision: 4 camera positions (2 external, 2 internal) with conduit and DVR room in utility.
- Intercom system at main gate.

5.7 Rainwater Harvesting (RWH) System

- Roof catchment area: ~250 sq.m; yield ~170,000 litres/year (Hyderabad avg. 800 mm rainfall).
- First-flush diverter to remove initial contaminated runoff.
- 15,000-litre underground RCC sump; filtered through gravel-sand filter.
- Overflow connected to recharge pit (2 m deep, 1 m dia with gravel fill) for groundwater recharge per GHMC mandate.

6. Work Breakdown Structure – Level 3

The WBS is structured at three levels: Level 1 = Project; Level 2 = Major work packages; Level 3 = Individual work activities / deliverables.

WBS Code	L1 – Project	L2 – Package	L3 – Activity / Deliverable
1	HOUSE	1.0 Site Prep	1.1 Site clearing & demolition
			1.2 Setting out & surveying
			1.3 Temporary facilities (site office, water, power)
2		2.0 Foundation	2.1 Excavation of column pits
			2.2 Excavation of plinth trenches
			2.3 PCC bed (1:4:8, 75 mm)
			2.4 Reinforcement fabrication for footings
			2.5 Concreting of isolated footings (M25)
			2.6 Column stub & plinth beam reinforcement
			2.7 Concreting of plinth beams
			2.8 Backfilling & compaction
3		3.0 Structural Frame	3.1 Reinforcement for ground floor columns
			3.2 Concreting of ground floor columns
			3.3 Reinforcement for beams
			3.4 Erection of beam & slab formwork
			3.5 Reinforcement for roof slab
			3.6 Concreting of roof slab (M25)
			3.7 Curing of slab (28 days water curing)
4		4.0 Masonry & Walls	4.1 External AAC block walls (200 mm)
			4.2 Internal partition walls (150 mm AAC)
			4.3 Lintels & window sills (RCC)
			4.4 Internal plastering (12 mm gypsum/cement)
			4.5 External plastering & weather coat
5		5.0 Roofing	5.1 Waterproofing screed (1:4 c-s)
			5.2 Cool roof reflective coating application
			5.3 PVC membrane waterproofing

			5.4 Roof parapet wall & coping
6		6.0 Flooring	6.1 Anti-termite treatment
			6.2 Base concrete for floors (1:2:4, 75 mm)
			6.3 Kota stone flooring (bedrooms, hallway)
			6.4 Engineered wood flooring (living, kids bed)
			6.5 Granite flooring (kitchen & toilets)
			6.6 Ceramic wall tiles (kitchen & bathroom walls)
7		7.0 Doors & Windows	7.1 Door frames (timber/UPVC) installation
			7.2 Window frames & glazing (aluminium)
			7.3 Door shutters & hardware
			7.4 Grills & safety screens
8		8.0 MEP Services	8.1 Electrical conduit & wiring (rough-in)
			8.2 Plumbing rough-in (concealed pipes)
			8.3 Drainage & sanitary fittings
			8.4 Electrical DB, MCBs & fixtures
			8.5 Solar water heater installation
			8.6 Fire detection system
9		9.0 Finishing	9.1 Interior painting (2 coats primer + 2 finish)
			9.2 Exterior painting (weather coat)
			9.3 False ceiling (gypsum board, wet areas)
			9.4 Kitchen cabinets & countertops
			9.5 Sanitary fixtures & plumbing connections
			9.6 Electrical fixtures, switches & sockets
10		10.0 External Works	10.1 Compound wall construction
			10.2 Gate fabrication & installation
			10.3 Driveway & open parking (paving)
			10.4 Rear private garden landscaping
			10.5 RWH sump construction & filtration setup
			10.6 Final cleanup & handover inspection

7. Quantity Calculations, Durations & Work Sequence

7.1 Assumptions & Dimensions

- Built-up area: 250 sq.m | External wall perimeter: ~67 m | Internal wall total length: ~40 m
- Floor height: 3.0 m | Slab thickness: 125 mm | Plinth height above GL: 0.6 m
- Number of isolated footings: 10 | Footing size: 1.5 m × 1.5 m | Footing depth: 1.0 m below GL
- Column size: 230 mm × 230 mm | Beam size: 230 mm wide × 450 mm deep (main), 230 × 300 (secondary)
- PCC bed: 75 mm thick (1:4:8) below all footings | Opening deduction: 15% of wall area for D&W

7.2 Quantity Calculations

#	Activity	Calculation	Quantity	Unit
A	Excavation – column pits	10 pits × 1.5×1.5×1.0 m	22.5	m ³
B	Excavation – plinth trenches	(67+40) m × 0.4 w × 0.6 d	25.7	m ³
	TOTAL EXCAVATION		48.2 ≈ 50	m ³
C	PCC bed (1:4:8)	10 × 1.5×1.5×0.075	1.7	m ³
D	Footing concrete (M25)	10 × 1.2×1.2×0.3 (net)	4.3	m ³
E	Column stub concrete	10 × 0.23×0.23×0.6	0.32	m ³
F	Plinth beam concrete	107 m × 0.23×0.30	7.4	m ³
G	Ground floor column concrete	10 × 0.23×0.23×3.0	1.6	m ³
H	Roof beam concrete (main+sec)	(67+40) m × 0.23×0.35	8.6	m ³
I	Roof slab concrete (M25)	250 × 0.125	31.25	m ³
	TOTAL CONCRETING		55.2	m ³
J	Footing steel (@ 80 kg/m ³)	4.3 × 80	344	kg
K	Column steel (@ 120 kg/m ³)	1.92 × 120	230	kg
L	Beam steel (@ 130 kg/m ³)	10.2 × 130	1,326	kg
M	Slab steel (@ 65 kg/m ³)	31.25 × 65	2,031	kg
	TOTAL STEEL		3,931 ≈ 4 MT	kg
N	External AAC wall masonry	67 m × 3.0 × 0.85 × 0.20	34.2	m ³
O	Internal AAC wall masonry	40 m × 3.0 × 0.85 × 0.15	15.3	m ³
	TOTAL MASONRY		49.5 ≈ 50	m ³
P	Internal plastering	(67+40) m × 3.0 m × 2 faces × 0.85	545	m ²
Q	External plastering	67 m × 3.0 m × 0.85	171	m ²

R	Kota stone flooring	250 – 40 (wood) – 30 (granite)	180	m ²
S	Engineered wood flooring	Living + Kids Bed (10×35 + 12×15 approx.)	40	m ²
T	Granite flooring (K & T)	Kitchen + 3 toilets	30	m ²
U	Ceramic tiles (walls – wet zones)	Bathroom + kitchen walls	120	m ²

7.3 Activity Durations – Productivity Basis

I D	Activity	Quantity	Productivity (unit/day)	Crew Size	Duration (days)	Predecessor
A	Site Clearing & Setting Out	1 LS	–	4	2	–
B	Excavation (all)	50 m ³	12 m ³ /day (JCB+4)	5	5	A
C	PCC Bed	1.7 m ³	4 m ³ /day	4	2	B
D	Footing Reinforcement	4.3 m ³ (steel fab)	2 m ³ /day	4	2	C
E	Footing Concreting	4.3 m ³	4 m ³ /day	4	2	D
F	Column Stub + Plinth Beam Reinf.	7.7 m ³	3 m ³ /day	4	3	E
G	Plinth Beam Concreting	7.4 m ³	4 m ³ /day	4	2	F
H	AAC Block Masonry (walls)	50 m ³	3 m ³ /day	4	17	G
I	Column Concrete (ground floor)	1.6 m ³	4 m ³ /day	4	3	G
J	Beam + Slab Formwork & Reinf.	40.1 m ³	5 m ³ /day	5	4	H,I
K	Roof Slab Concreting	31.25 m ³	8 m ³ /day (pump)	6	4	J
L	Plastering (internal + external)	716 m ²	80 m ² /day	4	9	K
M	Roofing (screed+coat+PVC)	250 m ²	65 m ² /day	4	4	K
N	Flooring (all types)	250 m ²	20 m ² /day	4	13	L
O	Electrical Works	1 LS	–	3	10	H
P	Plumbing (rough-in + finish)	1 LS	–	3	8	H
Q	Doors & Windows	1 LS	–	4	5	L
R	Painting (int + ext)	1 LS	–	4	7	L,O
S	Kitchen & Toilet Finishing	1 LS	–	3	6	P,N
T	External Works	1 LS	–	4	7	K

U	Cleanup & Handover	1 LS	–	4	2	R,S,Q,T, M
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Note: Total project duration = 73 working days (≈ 3.5 months excluding Sundays and holidays). Durations for MEP, painting, doors, and external works are assumed based on industry norms for a 250 sq.m. house.

8. Network Analysis – Activity on Node (AoN)

8.1 AoN Activity Table – ES, EF, LS, LF, TF, FF

All durations are in working days. Calculations based on forward pass and backward pass of CPM.

Critical Path: A → B → C → D → E → F → G → H → J → K → L → N → S → U = 73 days

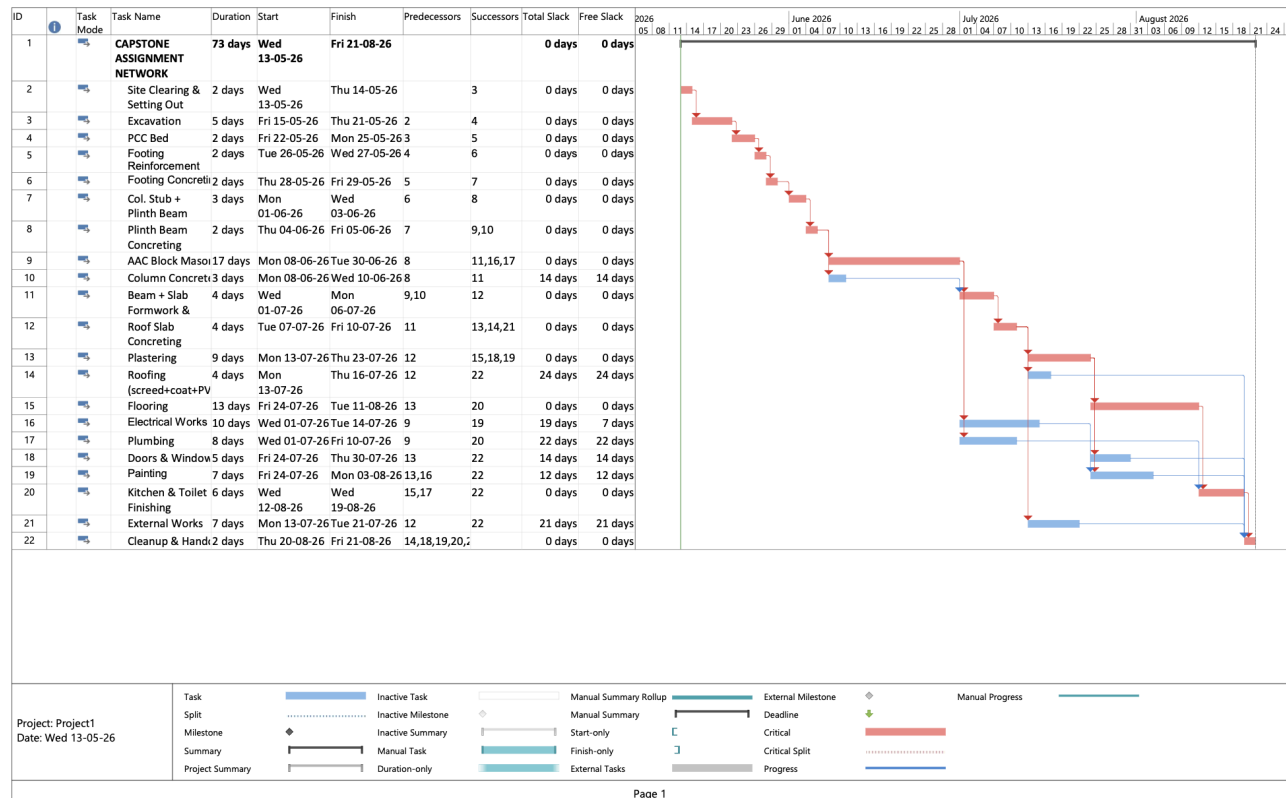
I D	Activity	Dur	ES	EF	LS	LF	TF	FF	Critical?	Pred.
A	Site Clearing & Setting Out	2	0	2	0	2	0	0	YES	–
B	Excavation	5	2	7	2	7	0	0	YES	A
C	PCC Bed	2	7	9	7	9	0	0	YES	B
D	Footing Reinforcement	2	9	11	9	11	0	0	YES	C
E	Footing Concreting	2	11	13	11	13	0	0	YES	D
F	Col. Stub + Plinth Beam Reinf.	3	13	16	13	16	0	0	YES	E
G	Plinth Beam Concreting	2	16	18	16	18	0	0	YES	F
H	AAC Block Masonry	17	18	35	18	35	0	0	YES	G
I	Column Concrete	3	18	21	32	35	14	14	no	G
J	Beam + Slab Formwork & Reinf.	4	35	39	35	39	0	0	YES	H,I
K	Roof Slab Concreting	4	39	43	39	43	0	0	YES	J
L	Plastering	9	43	52	43	52	0	0	YES	K
M	Roofing (screed+coat+PVC)	4	43	47	67	71	24	24	no	K
N	Flooring	13	52	65	52	65	0	0	YES	L
O	Electrical Works	10	18	28	54	64	36	24	no	H
P	Plumbing	8	18	26	57	65	39	39	no	H
Q	Doors & Windows	5	52	57	66	71	14	14	no	L
R	Painting	7	52	59	64	71	12	12	no	L,O
S	Kitchen & Toilet Finishing	6	65	71	65	71	0	0	YES	P,N

T	External Works	7	43	50	64	71	21	21	no	K
U	Cleanup & Handover	2	71	73	71	73	0	0	YES	R,S,Q,T,M

Legend: ES=Early Start | EF=Early Finish | LS=Late Start | LF=Late Finish | TF=Total Float | FF=Free Float | Red rows = Critical Path activities (TF = 0)

8.2 AoN Diagram – Space Reserved

Note: The AoN network diagram is prepared on MS Project. The diagram below.



9. Rate Analysis & Project Costs

Rate analysis is based on Hyderabad Schedule of Rates (DSR) 2024–25 and prevailing market rates. Labour rates: Skilled = ₹700/day, Unskilled = ₹500/day, Mason = ₹800/day.

9.1 Rate Analysis – Key Activities

A c t .	Activity	Qty	Unit	Rate (₹)	Amount (₹)	Basis
A	Site Clearing & Setting Out	1	LS	15,000	15,000	Labour + equipment
B	Excavation (earth, JCB)	50	m ³	1,440	72,000	JCB + labour + disposal
C	PCC 1:4:8 (75 mm)	1.7	m ³	3,200	5,440	Cement + agg + labour
D / E	Footing RCC M25	4.3	m ³	9,500	40,850	Cement+steel+shuttering+labe our
F / G	Plinth Beam RCC M25	7.4	m ³	9,500	70,300	Cement+steel+shuttering+labe our
H	AAC Block Masonry 200 mm	34.2	m ³	7,200	2,46,240	AAC blocks+mortar +labour
	AAC Block Masonry 150 mm	15.3	m ³	6,500	99,450	AAC blocks+mortar +labour
I	Column RCC M25	1.6	m ³	11,000	17,600	Cement+steel+shuttering
J - K	Beam + Slab RCC M25	39.85	m ³	10,500	4,18,425	Cement+steel+formwork+labe our
L	Plastering (int+ext)	716	m ²	200	1,43,200	Cement+sand +labour
M	Roofing system (coat+PVC)	250	m ²	650	1,62,500	Materials+labe our+equipment
N - R	Flooring – Kota Stone	180	m ²	550	99,000	Stone+adhesiv e+labour
N - S	Flooring – Eng. Wood	40	m ²	2,200	88,000	Wood+adhesiv e+labour

N	Flooring – Granite	30	m ²	1,400	42,000	Granite+labour
U	Ceramic tiles (walls)	120	m ²	800	96,000	Tiles+adhesive+labour
O	Electrical Works	1	LS	1,80,000	1,80,000	Wiring+conduit+DB+fixtures
P	Plumbing Works	1	LS	1,40,000	1,40,000	Pipes+ittings+fixtures
Q	Doors & Windows	1	LS	2,20,000	2,20,000	Frames+shutters+glazing
R	Painting (int+ext)	1	LS	1,20,000	1,20,000	Primer+paint+labour
S	Kitchen & Toilet Finishing	1	LS	1,80,000	1,80,000	Counters+fixtures+tiles
T	External Works	1	LS	90,000	90,000	Compd wall+gate+landscape
U	Cleanup & Handover	1	LS	20,000	20,000	Labour+skip hire

9.2 Project Cost Summary

Cost Head	Amount (₹)
Direct Construction Cost (from Rate Analysis above)	₹ 28,66,005
Add: Contingency @ 5%	₹ 1,43,300
Add: Site overheads & temp. works @ 5%	₹ 1,43,300
Sub-total (Civil Works)	₹ 31,52,605
Add: MEP Services (Electrical, Plumbing, HVAC)	₹ 4,20,000
Add: Kitchen & Toilet Finishing	₹ 1,80,000
Add: Doors, Windows & Hardware	₹ 2,20,000
Add: Painting & Polishing	₹ 1,20,000
TOTAL CONSTRUCTION COST	₹ 40,92,605
Verification: 250 sq.m × ₹16,400/sq.m ≈ ₹41,00,000	~₹ 41,00,000
NBC Basis: 2,690 sq.ft × ₹1,800/sq.ft	₹ 48,42,000
Estimated Project Budget (incl. premium finishes & RWH)	₹ 48L – ₹54L

Note: Rates are indicative based on Hyderabad DSR 2024-25. Final costs should be confirmed through competitive tendering. Budget estimate of ₹48L–₹54L from tutorial notes is adopted as the project budget.

10. Quality Assurance & Timely Completion Measures

10.1 Quality Assurance Measures

Quality will be controlled at material procurement, during construction, and at handover stages:

A. Concrete Quality Control:

- Weigh-batching of all concrete ingredients; no volume batching.
- Slump test at every pour; target slump 75–100 mm for columns/beams, 50–75 mm for slabs.
- Cube tests: 3 cubes per pour at 7 days and 3 cubes at 28 days; minimum 28-day strength $\geq 25 \text{ N/mm}^2$.
- Use of IS-marked cement (OPC 53); aggregates tested for grading, flakiness, and clay content.
- Minimum curing period: 14 days (columns, beams), 28 days (slabs). Use wet burlap or curing compound.

B. Masonry (AAC Block) Quality Control:

- Verify BIS certification (IS 2185-Part 3) of all AAC blocks at delivery; reject blocks with hairline cracks.
- Check block compressive strength (minimum 2.0 N/mm^2) via random sampling.
- Ensure all courses are level and plumb using a spirit level and a plumb-bob. Bed joint: 10–12 mm, wall joint: 8–10 mm.
- Mortar mix verification: 1:6 (cement:sand) by weight; test cube every 100 m^3 of masonry.

C. Steel Reinforcement Quality Control:

- Use Fe 415 / Fe 500D HYSD bars with IS 1786 certification; physical identification markings verified.
- Bend test and tensile test for each consignment per IS 1599.
- Bar schedules checked against drawings; cover blocks (50 mm footing, 40 mm columns/beams, 20 mm slabs) verified.
- No splicing at high-stress zones; lap lengths per IS 456 (min. $45 \times$ diameter).

D. Waterproofing & Roofing:

- Flood test on RCC slab: 50 mm water ponded for 24 hours – zero seepage acceptable.
- Cool roof coating applied in two coats; verify SRI > 78 from material data sheet.
- PVC membrane: lap joints $\geq 150 \text{ mm}$, heat-welded; visual inspection for bubbles or tears.

E. Third-Party Quality Checks:

- Independent laboratory testing for concrete cubes, soil tests, and steel samples.
- Structural engineer inspection at foundation, before slab pour, and at completion.
- GHMC inspections at plinth level, lintel level, and roof level (mandatory under Building Permit).

10.2 Timely Completion Measures

- Prepare a detailed 73-day schedule on MS Project; share with all subcontractors weekly.
- Conduct daily progress meetings (15 minutes) to identify and resolve any delays immediately.
- Maintain a material lookahead (2 weeks ahead): AAC blocks, steel, cement ordered before activity starts.
- Identify float activities (O, P, M, T) and use them as buffer for critical path recovery.
- Overlap non-dependent activities (e.g., Electrical rough-in H $\rightarrow$ O can start as soon as masonry above lintel is done).

- Weather contingency: concrete pours to be scheduled during clear weather; monsoon cover provided for critical operations.
- Penalty/incentive clause in subcontractor agreements to ensure timely performance.
- Track planned vs. actual S-curve (cost and physical progress) weekly; take corrective action if >5% deviation.
- Adequate resource levelling: avoid simultaneous deployment of all trades in same zone; sequence plastering zones.
- Final inspection checklist (100-point checklist) before handover; all snags to be closed within 7 days.

10.3 Regulatory Compliance Checklist

#	Compliance Item	Applicable Code	Status
1	GHMC Building Permit obtained before commencement	GHMC Building Rules 2017	Required
2	Setbacks maintained (front/rear/side) per GHMC	GHMC DCR	Required
3	Minimum room size 9.5 sq.m per NBC 2016 Part 3	NBC 2016 Pt.3	Compliant
4	Window area $\geq$ 1/6th of floor area (cross-ventilation)	NBC 2016 Pt.8	Compliant
5	RCC design per IS 456:2007 (M25, IS 1786 steel)	IS 456:2007	Compliant
6	Seismic design per IS 1893:2016 (Zone II, Hyderabad)	IS 1893:2016	Compliant
7	AAC blocks per IS 2185 Part 3	IS 2185:2008	Compliant
8	Electrical installation per IE Rules 2023	IE Rules 2023	Required
9	Rainwater harvesting as per GHMC RWH mandate	GHMC RWH Policy	Compliant
10	Labour welfare: PF, ESIC, safety equipment	Building & Other Construction Workers Act	Required